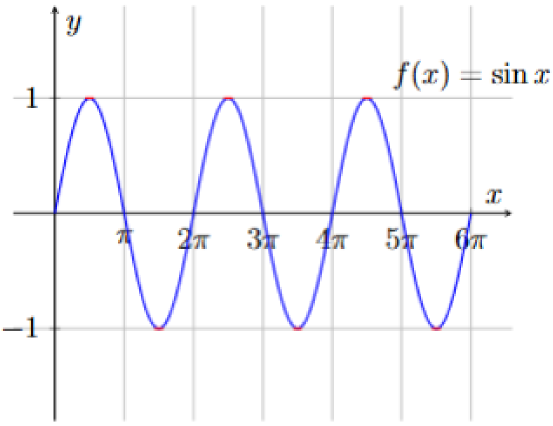
Lecture Four

1 Trigonometric Functions

1.1 Sine Function

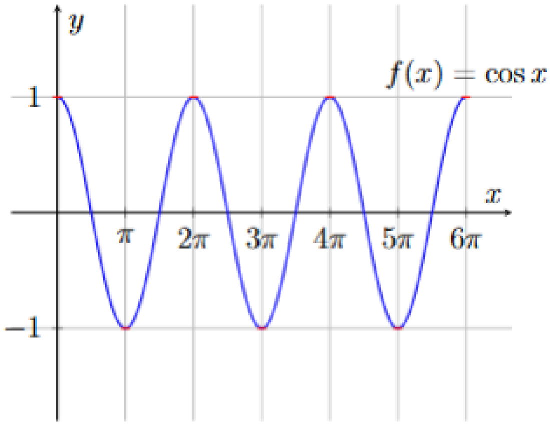


Properties of $y = \sin(x)$:

- $\forall x \in \mathbb{R},\ \sin(-x) = -\sin(x)$
- $\forall x \in \mathbb{R},\ \sin(x + 2\pi) = \sin(x)$
- $\forall k \in \mathbb{Z},\ \sin(x + 2k\pi) = \sin(x)$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$
- $\sin\left(\frac{\pi}{2} + x\right) = \cos(x)$

1.2 Cosine Function

$y = \cos x$ is the cosine function, and we know the domain $D = \mathbb{R}$ and the range $R = [-1, 1]$ and it is an even function.

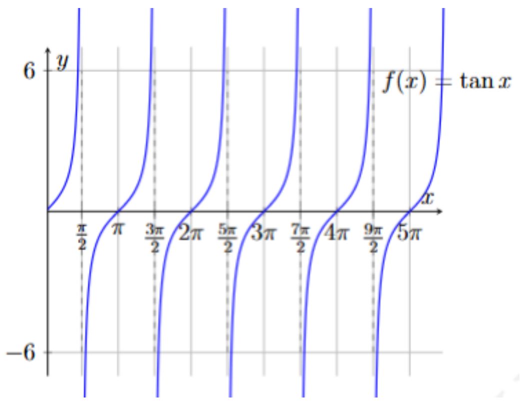


Properties of $y = \cos(x)$:

- $\forall x \in \mathbb{R},\ \cos(-x) = \cos x$
- $\forall x \in \mathbb{R},\ \cos(x + 2\pi) = \cos x$
- $\forall k \in I,\ \cos(x + 2k\pi) = \cos x$
- $\forall x \in \mathbb{R},\ \cos\left(\frac{\pi}{2} - x\right) = \sin x$
- $\forall x \in \mathbb{R},\ \cos\left(\frac{\pi}{2} + x\right) = -\sin x$

1.3 Tangent Function

$y = \tan x = \dfrac{\sin x}{\cos x}$ is the tangent function, we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R}$ and it is an odd function.

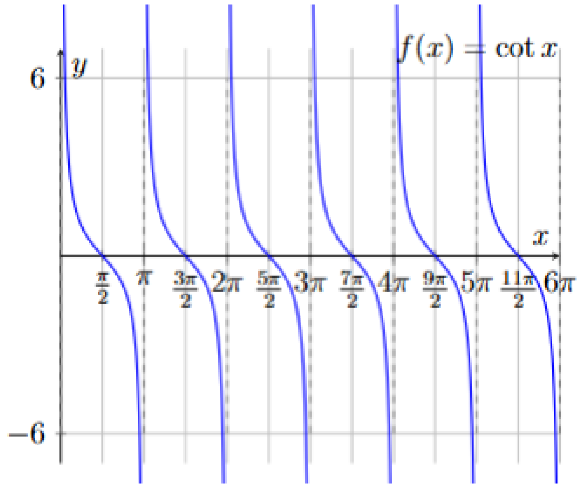


Properties of $y = \tan(x)$:

- $\forall x \in D,\ \tan(-x) = -\tan x$
- $\forall x \in D,\ \tan(x + 2\pi) = \tan x$
- $\forall k \in I, x \in D,\ \tan(x + 2k\pi) = \tan x$
- $\forall x \in D,\ \tan\left(\frac{\pi}{2} - x\right) = \cot x$
- $\forall x \in D,\ \cos\left(\frac{\pi}{2} + x\right) = -\cot x$

1.4 Cotangent Function

$y = \cot x = \dfrac{\cos x}{\sin x}$ is the cotangent function, and we know the domain $D = \mathbb{R} - \{k\pi, k \in I\}$ and the range $R = \mathbb{R}$ and it is an odd function.

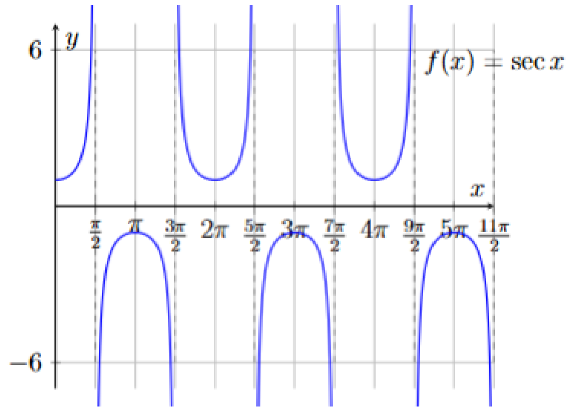


Properties of $y = \cot(x)$:

- $\forall x \in D,\ \cot(-x) = -\cot x$
- $\forall x \in D,\ \cot(x + 2\pi) = \cot x$
- $\forall k \in I, x \in D,\ \cot(x + 2k\pi) = \cot x$
- $\forall x \in D,\ \cot\left(\dfrac{\pi}{2} - x\right) = \tan x$
- $\forall x \in D,\ \cot\left(\dfrac{\pi}{2} + x\right) = -\tan x$

1.5 Secant Function

$y = \sec x = \dfrac{1}{\cos x}$ is the secant function, and we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R} - (-1, 1)$ and it is an even function.

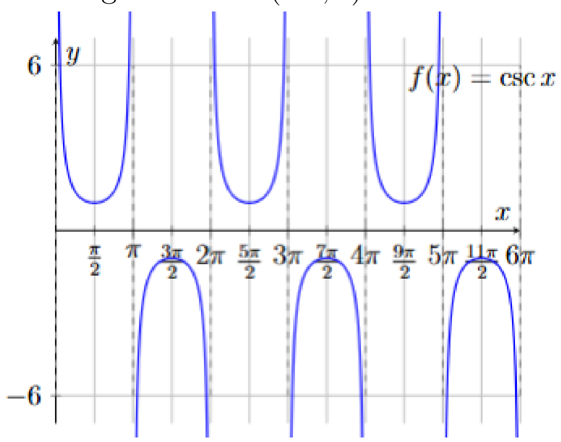


Properties of $y = \sec(x)$:

- $\forall x \in D,\ \sec(-x) = \sec x$
- $\forall x \in D,\ \sec(x + 2\pi) = \sec x$
- $\forall k \in I, x \in D,\ \sec(x + 2k\pi) = \sec x$
- $\forall x \in D,\ \sec\left(\dfrac{\pi}{2} - x\right) = \csc x$
- $\forall x \in D,\ \sec\left(\dfrac{\pi}{2} + x\right) = -\csc x$

1.6 Cosecant Function

$y = \csc x = \dfrac{1}{\sin x}$ is the cosecant function, we know the domain $D = \mathbb{R} - \left\{\dfrac{(2k-1)\pi}{2}, k \in I\right\}$ and the range $R = \mathbb{R} - (-1, 1)$ and it is an odd function.



Properties of $y = \csc(x)$:

- $\forall x \in D,\ \csc(-x) = -\csc x$
- $\forall x \in D,\ \csc(x + 2\pi) = \csc x$
- $\forall k \in I, x \in D,\ \sec(x + 2k\pi) = \csc x$
- $\forall x \in D,\ \csc\left(\dfrac{\pi}{2} - x\right) = \sec x$
- $\forall x \in D,\ \csc\left(\dfrac{\pi}{2} + x\right) = \sec x$

2 Differentiation of Trigonometric Functions

2.1 The Rules

1. $\frac{d}{dx}[\sin x] = \cos x$

2. $\frac{d}{dx}[\cos x] = -\sin x$

3. $\frac{d}{dx}[\tan x] = \sec^2 x$

Proof

$$\begin{aligned}\frac{d}{dx}[\tan x] = \frac{d}{dx}\left[\frac{\sin x}{\cos x}\right] &= \frac{\cos x \times \cos x - \sin x \times -\sin x}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

4. $\frac{d}{dx}[\cot x] = -\csc^2 x$

Proof

$$\begin{aligned}\frac{d}{dx}[\cot x] = \frac{d}{dx}\left[\frac{\cos x}{\sin x}\right] &= \frac{\sin x \times -\sin x - \cos x \times \cos x}{\sin^2 x} = -\frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\ &= -\frac{1}{\sin^2 x} = -\csc^2 x\end{aligned}$$

5. $\frac{d}{dx}[\sec x] = \sec x. \tan x$

Proof

$$\begin{aligned}\frac{d}{dx}[\sec x] = \frac{d}{dx}\left[\frac{1}{\cos x}\right] &= \frac{\cos x \times 0 - 1 \times -\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} \\ &= \frac{1}{\cos x}.\frac{\sin x}{\cos x} = \sec x. \tan x\end{aligned}$$

6. $\frac{d}{dx}[\csc x] = -\csc x. \cot x$

Proof

$$\begin{aligned}\frac{d}{dx}[\csc x] = \frac{d}{dx}\left[\frac{1}{\sin x}\right] &= \frac{\sin x \times 0 - 1 \times \cos x}{\sin^2 x} = -\frac{\cos x}{\sin^2 x} \\ &= -\frac{1}{\sin x}.\frac{\cos x}{\sin x} = -\csc x. \cot x\end{aligned}$$

In general:

No.	The Trig. Function	Derivative
(1)	$\frac{d}{dx}[\sin u(x)]$	$u^{'}(x)\cos u(x)$
(2)	$\frac{d}{dx}[\cos u(x)]$	$-u^{'}(x).\sin u(x)$
(3)	$\frac{d}{dx}[\tan u(x)]$	$u^{'}(x)\sec^2 u(x)$
(4)	$\frac{d}{dx}[\cot u(x)]$	$-u^{'}(x)\csc^2 u(x)$
(5)	$\frac{d}{dx}[\sec u(x)]$	$u^{'}(x)\sec u(x)\tan u(x)$
(6)	$\frac{d}{dx}[\csc u(x)]$	$-u^{'}(x)\csc u(x)\cot u(x)$

2.2 Examples

Example 2.2.1 *Find* $\frac{dy}{dx}$ *for the following functions:*

1. $y = 3\sin(2x^2+5) - \cos^3 x$

Sol

$$\frac{dy}{dx} = 3\times 4x \times \cos(2x^2+5) - 3\times\cos^2 x(-\sin x) = 12x\cos(2x^2+5) + 3\sin x\cos^2 x$$

2. $y = 4\sec(\cos x^2 + 1)$

Sol

$$\begin{aligned}\frac{dy}{dx} &= 4\left(-2x\sin x^2\right)\sec(\cos x^2+1)\tan(\cos x^2+1) \\ &= -8x\sin x^2\sec(\cos x^2+1)\tan(\cos x^2+1)\end{aligned}$$

3. $y = \sqrt{\tan(x\sec x)}$

Sol

Let $w = \tan(x\sec x) \Rightarrow y = \sqrt{w}$ *i.e.* $y = f(w)$. *And let* $u = x\sec x$ *i.e.* $u = h(x) \Rightarrow w = \tan u$ *i.e.* $w = g(u)$. *Hence*

$$\frac{dy}{dx} = \frac{dy}{dw}.\frac{dw}{du}.\frac{du}{dx}$$

Now: $\frac{dy}{dw} = \frac{1}{2\sqrt{w}}$, $\frac{dw}{du} = \sec^2 u$, $\frac{du}{dx} = x\sec x\tan x + \sec x$. *Therefore:*

$$\frac{dy}{dx} = \frac{1}{2\sqrt{w}}\times\sec^2 u\times(x\sec x\tan x+\sec x) = \frac{\sec^2(x\sec x)}{2\sqrt{\tan(x\sec x)}}.(x\sec x\tan x+\sec x)$$

4. $y = \left(\frac{\sin x}{\cot x + \tan x}\right)$

Sol

We can write $\dfrac{\sin x}{\cot x + \tan x}$ *as:*

$$\frac{\sin x}{\cot x + \tan x} = \frac{\sin x}{\dfrac{\cos x}{\sin x} + \dfrac{\sin x}{\cos x}} = \frac{\sin x}{\dfrac{\cos^2 x + \sin^2 x}{\sin x \cos x}} = \sin^2 x \cos x$$

Now $y = \csc(\sin^2 x \cos x)$, *let* $u = \sin^2 x \cos x \Rightarrow y = \csc u$ *and* $\dfrac{dy}{dx} = \dfrac{dy}{du} . \dfrac{du}{dx}$. *Therefore:*

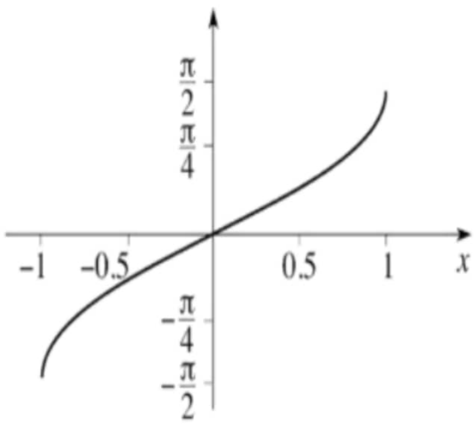
$$\frac{dy}{du} = -\csc u \cot u \quad , \frac{du}{dx} = -\sin^3 x + \cos x(2 \sin x \cos x) = \sin 2x \cos x - \sin^3 x$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= -\csc u \cot u \times (\sin 2x \cos x - \sin^3 x) \\ &= -(\sin 2x \cos x - \sin^3 x) \csc(\sin^2 x \cos x) \cot(\sin^2 x \cos x) \end{aligned}$$

3 The Inverse Trigonometric Functions

3.1 The Inverse Sine Function

Let $x = \sin y$, then the inverse sine function is $y = \arcsin(x) = \sin^{-1} x$, the domain of definition is $D = [-1, 1]$, the range $R = \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$, and it is an odd function, where $\sin^{-1}(-x) = -\sin^{-1} x$.



The derivative of inverse sin function

$$y = \sin^{-1} x \Rightarrow x = \sin y \Rightarrow 1 = \cos y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$$

And we know that $\cos y = \pm\sqrt{1 - \sin^2 y}$. Since $y \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right] \Rightarrow \cos y > 0$, hence

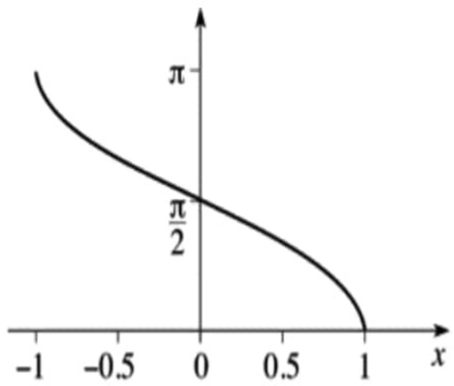
$$\cos y = \sqrt{1 - \sin^2 y}$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}}$$

3.2 The Inverse Cosine Function

Let $x = \cos y$, then the inverse cosine function is $y = \arccos(x) = \cos^{-1} x$, the domain of definition is $D = [-1, 1]$, the range $R = [0, \pi]$, and it is an odd function, where $\cos^{-1}(-x) = \cos^{-1} x$.



The derivative of inverse cos function

$$y = \cos^{-1} x \Rightarrow x = \cos y \Rightarrow 1 = -\sin y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\sin y}$$

And we know that $\sin y = \pm\sqrt{1 - \cos^2 y}$. Since $y \in [0, \pi] \Rightarrow \sin y > 0$, hence

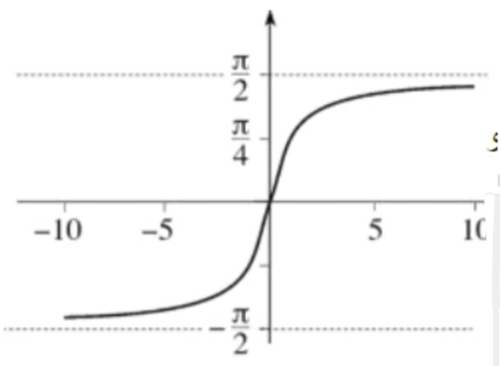
$$\sin y = \sqrt{1 - \cos^2 y}$$

Therefore

$$\boxed{\frac{dy}{dx} = \frac{-1}{\sqrt{1 - x^2}}}$$

3.3 The Inverse Tangent Function

Let $x = \tan y$, then the inverse tangent function is $y = \arctan(x) = \tan^{-1} x$, the domain of definition is $D = \mathbb{R}$, the range $R = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and it is an odd function, where $\tan^{-1}(-x) = -\tan^{-1} x$.



The derivative of inverse tan function

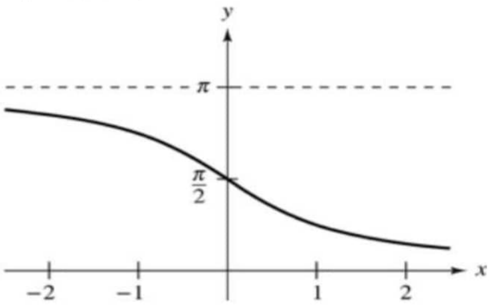
$$y = \tan^{-1} x \Rightarrow x = \tan y \Rightarrow 1 = \sec^2 y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

We know $\sec^2 y = 1 + \tan^2 y = 1 + x^2$

$$\therefore \boxed{\frac{dy}{dx} = \frac{1}{1 + x^2}}$$

3.4 The Inverse Cotangent Function

Let $x = \cot y$, then the inverse tangent function is $y = \cot^{-1} x$, the domain of definition is $D = \mathbb{R}$, the range $R = (0, \pi)$, and it is an odd function, where $\cot^{-1}(-x) = -\cot^{-1} x$.



The derivative of inverse cot function

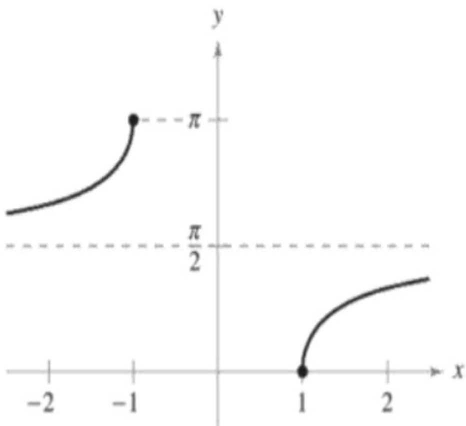
$$y = \cot^{-1} x \Rightarrow x = \cot y \Rightarrow 1 = -\csc^2 y \, \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\csc^2 y}$$

We know $\csc^2 y = 1 + \cot^2 y = 1 + x^2$

$$\therefore \boxed{\frac{dy}{dx} = \frac{-1}{1+x^2}}$$

3.5 The Inverse Secant Function

Let $x = \sec y$, then the inverse cosine function is $y = \sec^{-1} x$, the domain of definition is $D = \mathbb{R} - (-1, 1)$, the range $R = \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$, and it is an even function, where $\sec^{-1}(-x) = \sec^{-1} x$.



The derivative of inverse sec function

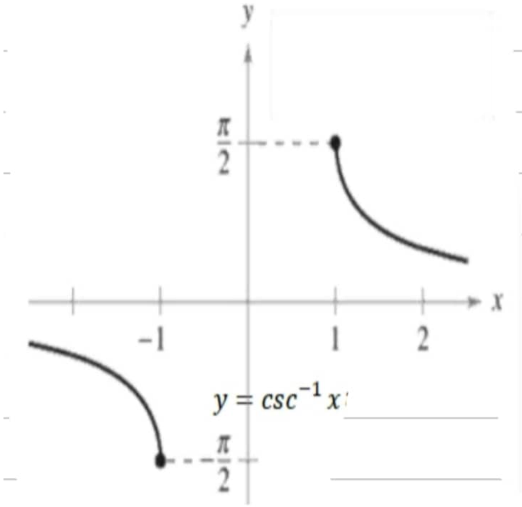
$$y = \sec^{-1} x \Rightarrow x = \sec y \Rightarrow 1 = \sec y \, \tan y \, \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\sec y \, \tan y}$$

And we know $\tan y = \pm\sqrt{\sec^2 y - 1}$, either $y \in \left[0, \frac{\pi}{2}\right)$ this implies that $\tan y > 0$ and we take $\tan y = \sqrt{\sec^2 y - 1}$ and thus $\sec y \, \tan y = |\sec y|\sqrt{\sec^2 y - 1} > 0$ or $y \in \left(\frac{\pi}{2}, \pi\right]$, which is implies that $\tan y$ & $\sec y < 0 \Rightarrow \tan y = -\sqrt{\sec^2 y - 1}$ and thus $\sec y \tan y = |\sec y|\sqrt{\sec^2 y - 1} > 0$. Now

$$\frac{dy}{dx} = \frac{1}{|\sec y|\sqrt{\sec^2 y - 1}} \Rightarrow \boxed{\frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2 - 1}}}$$

3.6 The Inverse Cosecant Function

Let $x = \csc y$, then the inverse cosine function is $y = \csc^{-1} x$, the domain of definition is $D = \mathbb{R} - (-1, 1)$, the range $R = \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$, and it is an odd function, where $\csc^{-1}(-x) = -\csc^{-1} x$.



The derivative of inverse csc function

$$y = \csc^{-1} x \Rightarrow x = \csc y \Rightarrow 1 = -\csc y\ \cot y\ \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{-1}{\csc y\ \cot y}$$

And we know $\cot y = \pm\sqrt{\csc^2 y - 1}$, either $y \in \left[-\frac{\pi}{2}, 0\right)$ this implies that $\csc y > 0$ and we take $\cot y = -\sqrt{\sec^2 y - 1}$ and thus $\csc y\ \cot y = |\csc y|\sqrt{\csc^2 y - 1} > 0$ or $y \in \left(0, \frac{\pi}{2}\right]$, which is implies that $\csc y > 0 \Rightarrow\ \cot y = \sqrt{\csc^2 y - 1}$ and thus $\csc y \cot y = |\csc y|\sqrt{\csc^2 y - 1} > 0$. Now

$$\frac{dy}{dx} = \frac{-1}{|\csc y|\sqrt{\csc^2 y - 1}} \Rightarrow \boxed{\frac{dy}{dx} = \frac{-1}{|x|\sqrt{x^2 - 1}}}$$

Example 3.6.2 *Find $\frac{dy}{dx}$ for the following functions:*

1. $y = \frac{x}{2}\sin^{-1} 2x$

Sol

$$\frac{dy}{dx} = \frac{x}{2}.\frac{2}{\sqrt{1 - 4x^2}} + \frac{1}{2}\sin^{-1} 2x = \frac{x}{\sqrt{1 - 4x^2}} + \frac{1}{2}\sin^{-1} 2x$$

2. $y = \cos^{-1}(x \cos x)$

Sol

Let $u = x\cos x \Rightarrow u^{'}(x) = \cos x - x\sin x$ and we know $\frac{d}{dx}\left(\cos^{-1}(u(x))\right) = \frac{-u^{'}(x)}{\sqrt{1 - (u(x))^2}}$

$$\therefore \frac{dy}{dx} = \frac{-(\cos x - x\sin x)}{\sqrt{1 - (x\cos(x))^2}} = \frac{x\sin x - \cos x}{\sqrt{1 - x^2\cos^2 x}} \qquad \square$$

3. $y = \frac{\tan^{-1} x}{x^2 + 1}$

Sol

$$\frac{dy}{dx} = \left(\frac{(x^2+1) \times \dfrac{1}{x^2+1} - \tan^{-1} x \times (2x)}{(x^2+1)^2} \right) = \frac{1 - 2x \tan^{-1} x}{(x^2+1)^2}$$

4. $y = \tan^{-1}\left(\dfrac{\tan x + 1}{\tan x - 1}\right)$

Sol

We know that $\dfrac{d}{dx}\left[\tan^{-1}(u(x))\right] = \dfrac{u'(x)}{1+(u(x))^2}$

$$\begin{aligned}
\therefore \frac{dy}{dx} &= \frac{1}{1+\left(\dfrac{\tan x+1}{\tan x-1}\right)^2} \times \frac{(\tan x - 1)\times \sec^2 x - (\tan x + 1)\times \sec^2 x}{(\tan x - 1)^2} \\
&= \left(\frac{(\tan x-1)^2}{(\tan x+1)^2 + (\tan x - 1)^2}\right) \times \frac{\sec^2 x[\tan x - 1 - \tan x - 1]}{(\tan x - 1)^2} \\
&= \frac{-2\sec^2 x}{2(\tan^2 x + 1)} = \frac{-2\sec^2 x}{2\sec^2 x} = -1
\end{aligned}$$

Another Solution: *We can write* $\dfrac{\tan x + 1}{\tan x - 1}$ *as:*

$$\begin{aligned}
\frac{\tan x + 1}{\tan x - 1} &= \frac{\dfrac{\sin x}{\cos x} + 1}{\dfrac{\sin x}{\cos x} - 1} = \frac{\sin x + \cos x}{\sin x - \cos x} = \frac{\sin^2 x + 2x \sin x \cos x + \cos^2 x}{\sin^2 x - \cos^2 x} \\
&= -\left[\frac{1+\sin 2x}{\cos 2x}\right] = -[\sec 2x + \tan 2x]
\end{aligned}$$

Now, $\tan^{-1}\left(\dfrac{\tan x + 1}{\tan x - 1}\right) = -\tan^{-1}[\sec 2x + \tan 2x]$

$$\begin{aligned}
\therefore \frac{dy}{dx} &= -\left(\frac{2\sec 2x \tan 2x + 2\sec^2 x}{1 + (\sec 2x + \tan 2x)^2}\right) = -\left(\frac{2\sec 2x \tan 2x + 2\sec^2 x}{1+\sec^2 x + 2\sec 2x \tan 2x + \tan^2 x}\right) \\
&= -\left(\frac{2\sec 2x \tan 2x + 2\sec^2 x}{2\sec 2x \tan 2x + 2\sec^2 x}\right) = -1.
\end{aligned}$$

Another Solution:

$$\begin{aligned}
y = \tan^{-1}\left(\frac{\tan x + 1}{\tan x - 1}\right) &= -\tan^{-1}\left(\frac{\tan x + 1}{1 - \tan x}\right) = -\tan^{-1}\left(\frac{\tan x + \tan\dfrac{\pi}{4}}{1 - \tan x \times \tan\dfrac{\pi}{4}}\right) \\
&= -\tan^{-1}\left(\tan\left(x + \frac{\pi}{4}\right)\right) = -\left(x + \frac{\pi}{4}\right) \Rightarrow \boxed{y = -x - \frac{\pi}{4} \Rightarrow \frac{dy}{dx} = -1}
\end{aligned}$$

Exercise 3.6.1 *Find* $\dfrac{dy}{dx}$ *for the following functions.*

1.

$$y = 2\sin^{-1}(x-1)$$

Sol

2.

$$y = \tan^{-1}\left(\frac{x}{a}\right) + 2\sin^{-1}\left(\frac{a}{x}\right)$$

Sol

3.

$$y = \frac{\cos^{-1} x^2}{x+1}$$

Sol

4.

$$y = e^{x^2} \sin^{-1} x$$

Sol

5.

$$y = \sin^{-1} 2x + \cos^{-1} 2x$$

Sol

6.

$$y = x\cos^{-1} x - \sqrt{1-x^2}$$

Sol

7.

$$y = \frac{1}{2}\left[x\sqrt{4-x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right]$$

Sol

8.

$$y = \cos x . \sec^{-1} x$$

Sol

9.

$$y = e^{\sin^{-1} x} + \frac{1}{\sqrt{1+x^2}}$$

Sol

10.

$$y = \csc\left[\frac{x}{x+1}\right] - \sec^{-1}\left[x + \frac{1}{x}\right]$$

Sol

11.

$$y = \frac{\cot^{-1} x}{1 + x^2}$$

Sol

12.

$$y = \ln\left(x - \tan x\right)$$

Sol

13.

$$y = \csc^{-1}\left[\frac{1 - x}{x + 1}\right]$$

Sol

14.

$$y = \ln\sqrt{\frac{x + 1}{x - 1}} + \tan^{-1} 2x$$

Sol